KHAWAJA FAREED UNIVERSITY OF ENGINEERING & INFORMATION TECHNOLOGY

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P R O J E C T R E P O R T

Semester Project in Flutter

Submitted By

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Submitted To

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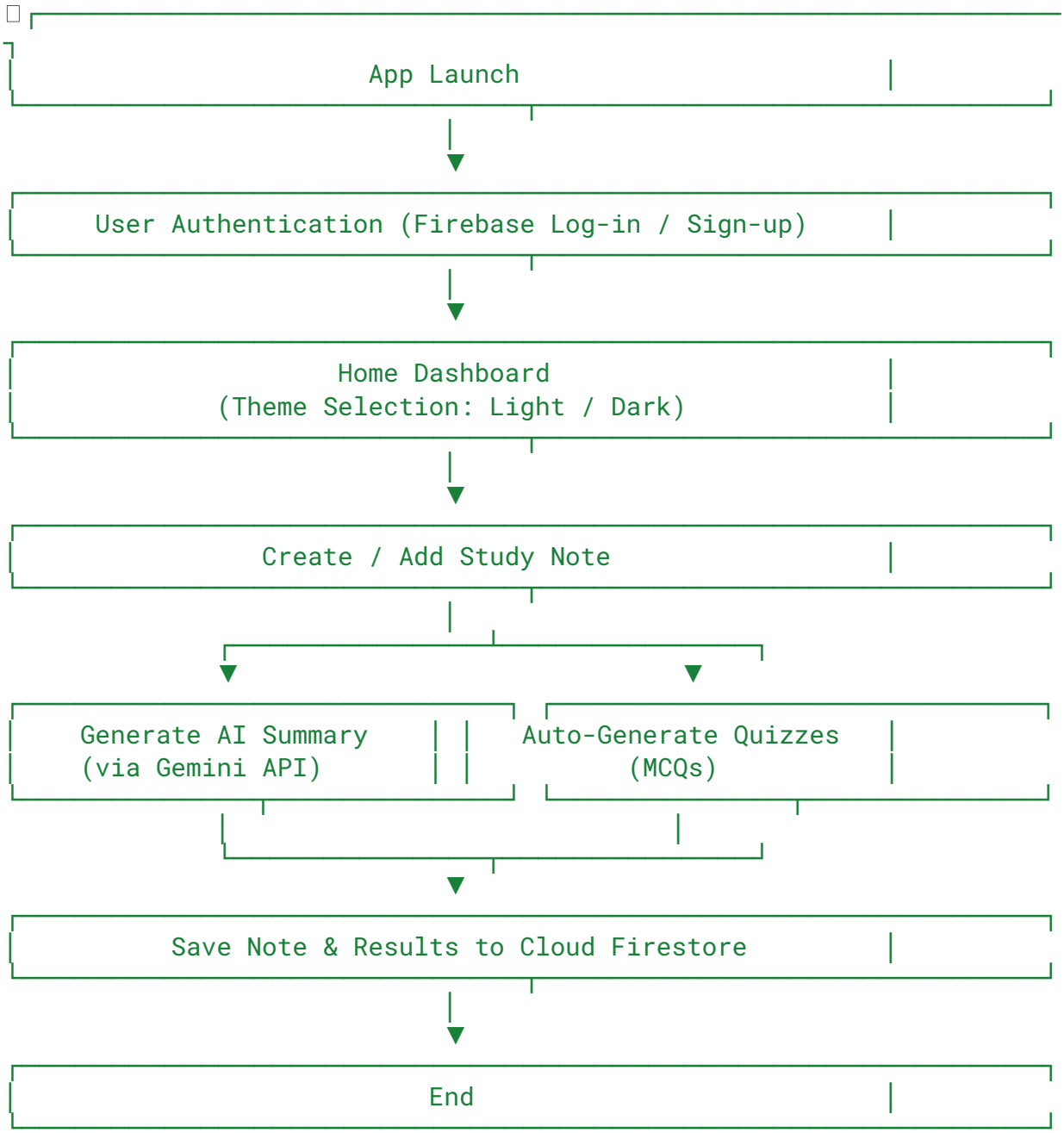
Project Title

Learnozi - AI Study Companion

Short description:

A Flutter-based AI study app that helps students summarize text notes and generate automatic quizzes using AI. It provides a home dashboard, text note saving functionality, AI summaries, and a smooth interface with light and dark themes on Android and iOS.

System Workflow



Functional Requirements

Core Features (AI & Notes)

- **FR-01:** The system shall provide a Home Dashboard for students to view saved notes.
- **FR-02:** The system shall allow users to securely log in using Firebase Authentication.
- **FR-03:** The system shall allow users to type and save text-based study notes.
- **FR-04:** The system shall generate simplified summaries of saved notes using AI.
- **FR-05:** The system shall auto-generate basic MCQs from the saved text.

Theme & User Experience

- **FR-06:** The system shall provide a Light Theme.
- **FR-07:** The system shall provide a Dark Theme.
- **FR-08:** The system shall allow users to switch between the two themes.
- **FR-09:** The system shall provide a smooth and intuitive user interface.

Platform & Code Quality

- **FR-10:** The application shall support Android.
- **FR-11:** The project shall use Dart null safety.

Suggested Technology

Component	Technology / Purpose
Frontend	Flutter + Dart
Backend & Database	Firebase (Authentication & Cloud Firestore)
AI Integration	Google Gemini API (via HTTP)
Platforms	Android
Theme	Light Theme + Dark Theme
Code Safety	Dart null safety

Main User Flow

1. User launches the Learnozi App.
2. User logs in via Firebase.
3. User adds a new study note (text).
4. App fetches an AI summary and displays it.
5. User takes a short auto-generated quiz based on their note.
6. User can switch between Light and Dark themes.

Feature Summary

Feature	Description
Dashboard	Displays recent study notes.
Firebase Auth	Secure user login system.
Cloud Firestore	Saves notes and summaries in real-time.
AI Summarizer	Processes text into easy-to-read points.
Auto-Quiz	Tests student knowledge via AI-generated MCQs.
Dual Themes	Provides Light Theme and Dark Theme.
Platforms	Designed for Android.

Project Overview

The Learnozi Flutter App is an intelligent, simple mobile application for modern students. It saves study notes using Firebase Cloud Firestore, handles AI summarization via Gemini, and presents study materials through a smooth, intuitive interface. Users can explore summaries, take short quizzes, and use a Light or Dark theme, with a seamless experience on Android.